



Capstone Project Phase B

Adaptive Virtual Reality Museum

A Closed-Loop Highlight Intervention Study in Cultural-Heritage VR

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Project repository: github.com/JosephDamouni/Project_Phase-A

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1. Problem Overview and Background

Virtual reality has become a powerful medium for cultural heritage, allowing sites such as the Berat Ethnographic Museum in Albania - a nineteenth-century Ottoman house museum in a UNESCO World Heritage city - to be experienced by people who cannot visit in person. Recent reviews document the growing use of immersive VR for cultural-heritage reconstruction and museum exhibitions [12, 13]. Most VR museums, however, inherit a static delivery paradigm from physical exhibitions: content is authored in advance and presented identically to every visitor. Phase A of this project identified the resulting mismatch as its core problem: engagement varies continuously in immersive environments [5], so static presentation causes cognitive overload during rapid movement and under-stimulation during attentive dwelling. Modern standalone headsets, meanwhile, stream rich behavioural signals - where the controller ray points, how the head moves, how fast the visitor walks, where the hands are - that could reveal exactly when a visitor is losing interest, without questionnaires, eye trackers, or physiological sensors.

Phase A therefore specified an engagement-aware VR museum: the requirements, system architecture, and UML design for a platform that reconstructs the museum faithfully, infers the visitor's engagement in real time from these implicit signals through a transparent rule-based classifier, and closes the loop by adapting the experience to the inferred state. Phase A also defined the research programme and its success criteria; full implementation and evaluation were deferred to Phase B.

Two research strands frame the work. First, adaptive VR systems - including the closed-loop framework published from this project [1, 2] - have largely adapted content, for example exhibit-text depth, to inferred user state, without intervening on visual attention directly. Second, the attention-guidance literature studies the cue mechanism itself: guidance taxonomies for cinematic VR [14], saliency and brightness modulation [6, 16], subtle gaze direction [3], comparisons of cue types and placements in cultural-heritage settings [7, 8], gaze-based interaction and coordination [9, 15], and peripheral versus central cues evaluated on presence and workload [17]. Guidance mechanisms have been shown to reduce cognitive load and improve learning in immersive environments [18], but they are almost always deployed as fixed features, present throughout a session. Across this entire body of work, cue timing is treated as a property of the authored experience or of the experimental protocol - not as a closed-loop decision driven by the visitor's moment-to-moment state [11].

This gap defines the final research question of the project. Holding the cue's appearance, its target-selection rule, and the exhibit content constant, the project asks: (RQ1) does engagement-driven cue timing improve exploration and reading engagement relative to fixed-schedule timing; and (RQ2) if a difference is observed, is it attributable to highlight dose - one policy simply firing more cues - or to timing, cues arriving at moments when the visitor can benefit from them? The research was conducted in two iterations: an initial study of engagement-adaptive text pacing, whose results did not provide sufficiently meaningful insight into visitor engagement, and the redesigned highlight-timing study reported in this booklet (Sections 3 and 5).

2. Description of the Project

2.1 General Description

The project delivers a photorealistic six-room VR reconstruction of the Berat Ethnographic Museum for the Meta Quest 3, instrumented as a research platform for engagement-aware interaction. Its objectives are to (i) reconstruct the museum faithfully in collaboration with a cultural-heritage partner; (ii) infer a five-level engagement state on-device every two seconds from implicit behavioural signals; (iii) deliver a transient perceptual highlight - a glow on a nearby unexamined artefact - under two interchangeable trigger policies; (iv) compare those policies in a controlled two-group experiment; and (v) log complete behavioural telemetry to Firebase Firestore for offline analysis.

The system serves three kinds of users. Museum visitors - the remote audience of the physical museum - explore the rooms freely with full 6DoF locomotion, open localised information panels on every artefact, and are gently guided by the highlight cue; all content is available in English and Albanian (Shqip), switched at runtime through Unity string tables. The cultural-heritage partner gains an accessible, faithful digital twin of the collection. Researchers gain an auditable experimental platform in which every session's configuration, allocation, and behavioural record is recoverable from the telemetry.

A gamification layer motivates exploration identically in both experimental conditions: eighteen hidden collectible cards (three per room) open discovery panels with localised cultural descriptions; thirteen achievement badges reward milestones (one, three, six, nine, twelve, and eighteen cards), per-room completion, and full completion; and a score accumulates one hundred points per card, ten per artefact examined, and a one-thousand-point master-collector bonus. An end-of-session achievements screen summarises badges, cards, per-room time, and score in the participant's language.



Figure 1. The reconstructed main guest room in the VR application.

2.2 High-Level Architecture

The system is a Unity 6 (URP) application organised into six logical layers (Table 1, Figure 2). Layers communicate through MonoBehaviour singleton references and event callbacks rather than dependency injection - a deliberate choice that lets the experiment configuration enable or disable whole subsystems at runtime: at launch,

ExperimentConditionManager allocates the participant to a condition and enables exactly one trigger policy, while a master switch retires the avatar, speech-bubble, and adaptive-text surfaces in both conditions. A single SessionPhaseManager gates both policies during the 40-second acclimation window, so neither condition can fire while the participant is still orienting.

Layer	Main components	Responsibility
Sensing	ControllerGazeTracker, HeadTrackingAnalyzer, HandProximityTracker, ReadingBehaviorTracker	Continuous behavioural-signal acquisition, normalised to [0, 1]
Classification	EngagementClassifier	Signal fusion, smoothing, five-level engagement inference every 2 s
Intervention	MuseumAgentController (Adaptive), NonAdaptiveHighlightPolicy (Control), AgentAttentionDirector, GuidedHighlight	Engagement-driven or fixed-schedule highlight delivery over a shared actuator
Content	ArtifactInteractionHandler, LLMAdaptiveContentManager	Artefact panels; LLM text with fixed, engagement-independent settings
Gamification	HiddenCard, BadgeManager, ScoreManager, AchievementsManager	Cards, badges, scoring - identical across conditions
Persistence	FirebaseLogger, FirebaseManager, SessionPhaseManager, EndOfSessionSummary	Telemetry, session management, acclimation gate, aggregation

Table 1. The six architectural layers.

The sensing layer is a set of singletons, each on its own update schedule, exposing a normalised signal the classifier reads at each tick. ControllerGazeTracker, the primary channel, polls the XR ray interactor every frame, accumulating dwell while the ray rests on a tagged interactive object and decaying it after a grace period once the ray moves away. HeadTrackingAnalyzer samples the XR rig at half-second intervals, deriving head angular velocity and locomotion speed. HandProximityTracker rescans interactive objects every five seconds using the hand-tracking subsystem (controller position as fallback) and computes an inverse-distance proximity score. ReadingBehaviorTracker is event-driven, emitting a per-panel reading score from open/close events that reflects view duration, a wasLikelyRead flag, and recency.

Deployment is deliberately simple (Figure 3): everything runs on the headset. The Unity client performs all sensing, classification, and intervention locally; Firebase Firestore receives telemetry over Wi-Fi (writes are queued while offline and replayed on reconnection); and the OpenAI API serves exhibit text generated under fixed, engagement-independent settings identical in both conditions [19]. No dedicated server is required, which keeps study logistics to a headset and a Wi-Fi connection.

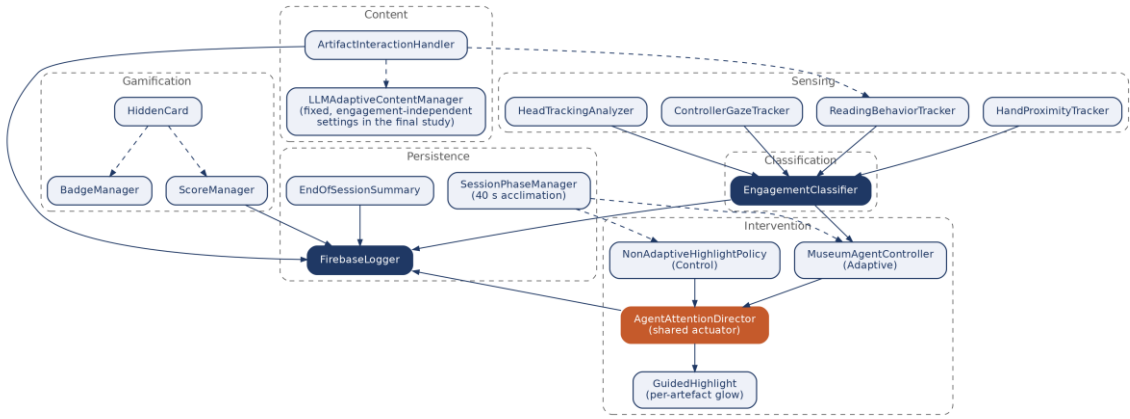


Figure 2. Package-level view: core components by layer. Solid arrows show primary data flow; dashed arrows secondary dependencies.

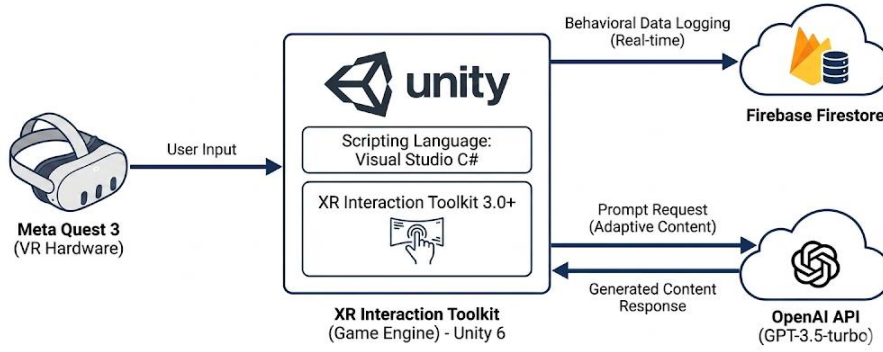


Figure 3. Deployment view: Quest 3 client, Firebase Firestore telemetry, and the OpenAI API.

2.3 Engagement Classification

EngagementClassifier executes a five-stage pipeline every two seconds (Figure 4); all weights, thresholds, and constants below were fixed during development, before participant recruitment, and were never adjusted in response to participant outcomes.

Stage 1 - physical signals. The classifier polls the four sensors and first checks tracking validity: if head displacement stays below 0.0001 m for more than ten seconds (a headset set down, or a stale pose), the session is flagged tracking-invalid, sensors are neutralised to mid-range values, and Firebase writes are suppressed. Movement-derived engagement signals are interpreted cautiously because locomotion and cognitive load can interact in VR [10]. The physical composite is $S_{phys} = \text{gaze} \times 0.40 + \text{head} \times 0.25 + \text{movement} \times 0.25 + \text{hand} \times 0.10$, each signal clamped to [0, 1]. Gaze carries the largest weight because it is the only channel under direct, intentional control in a laser-pointer interaction style; hand proximity carries the smallest because controllers make it partially redundant.

Stage 2 - reading verification. The event-driven reading score is admitted only when corroborated: a content-anchored check zeroes it whenever no panel is open or recently closed (≤ 15 s), so tracker false-positives near text-bearing surfaces cannot inflate engagement. Panels held under two seconds with `wasLikelyRead = false` ('skipped opens') do not extend the verification window.

Stage 3 - gaze-floor correction. An open content panel occludes the artefact's collider, so the controller raycast falsely reports zero gaze in the middle of the system's strongest engagement evidence. As an isolated, sensor-level correction, the gaze channel is floored to 0.85 while a panel is open and 0.70 during sustained high-confidence reading. Reading still contributes to the final score only through its own 25% term - the floor repairs the broken sensor, it does not double-count reading.

Stage 4 - weighted fusion. The channels combine as $E_{\text{raw}} = 0.75 \times S_{\text{phys}} + 0.25 \times S_{\text{read}}$, the reading weight scaled by its confidence and any unused reading weight redistributed to the physical term, so without verified reading the score reduces exactly to S_{phys} . The 75/25 split reflects coverage: physical signals are continuous, reading evidence exists only around panel interactions.

Stage 5 - smoothing, levels, gates. The raw score passes through an asymmetric exponential moving average ($\alpha = 0.35$ on ascent, 0.525 on descent), so genuine drops register promptly while upward flickers are dampened. The smoothed score maps to five ordinal levels (Table 2), and two velocity gates then cap the level at Disengaged when the participant moves at ≥ 0.8 m/s (soft) or ≥ 1.5 m/s (strict) without reading evidence - both suppressed during a short reading-grace window so a visitor walking away from a just-read panel is not penalised. Inference latency is sub-millisecond on the Quest 3.

Engagement level	Smoothed score	Interpretation
Highly Engaged	≥ 0.75	Sustained focused inspection / reading
Engaged	0.55 – 0.75	Active attention to exhibits
Neutral	0.35 – 0.55	Ordinary exploration
Disengaged	0.20 – 0.35	Reduced attention; intervention candidate
Highly Disengaged	< 0.20	Sustained disengagement; intervention candidate

Table 2. Engagement-level thresholds.

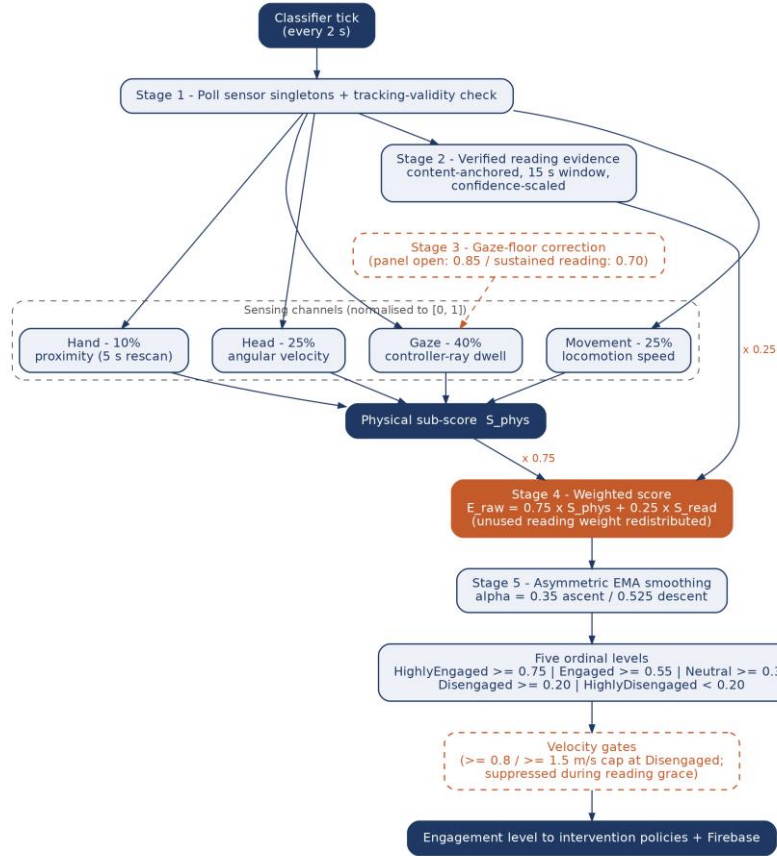


Figure 4. The five-stage engagement-classification pipeline with the implemented signal weights.

2.4 Highlight Intervention

Shared actuator. Both conditions call one component, AgentAttentionDirector, which isolates when a cue fires from where it lands. On each trigger request it first checks the global cooldown and the fifteen-highlight session cap. If clear, it scores every artefact within 0.6–6.0 m of the participant's head - +100 if not yet examined, +50 if in the current room, -5 per metre of distance - and highlights the top candidate: a pulsing child-GameObject glow with an optional spatialised chime, lasting four to fifteen seconds, dismissable early by sustained gaze (head-direction dot ≥ 0.85 held 1.5 s). A fifteen-second follow-through window then records whether the participant opened the cued artefact's panel - the primary behavioural response - plus gaze follow-through and latency-to-first-interaction, and a snapshot twenty seconds after the fire captures the smoothed-engagement delta underlying hypothesis H3.

Adaptive policy. MuseumAgentController runs a state machine over the classifier's updates (Figure 5). Nothing beyond idle is permitted during the 40-second acclimation. When Disengaged or Highly Disengaged is observed, a five-second hysteresis timer must validate the state - transient dips do not qualify. A highlight fires only after the validated state has been held for at least ten further seconds and a content-recency guard confirms no panel was opened moments before (the visitor may be between exhibits, not disengaged). The policy deliberately does not fire on sustained Neutral engagement, keeping the manipulation focused on disengagement recovery. The logged triggerReason records the validated state-machine category, not the instantaneous reading at fire time.

Control policy. NonAdaptiveHighlightPolicy is intentionally simple (Figure 6): on room entry a dwell timer starts, and after fifteen continuous seconds in the room (post-acclimation) it requests a highlight with reason fixed_room_dwell. It imposes no per-room cap; its cue rate is bounded by exactly the same global cooldown and session cap as the Adaptive policy, so the two conditions differ only in when cues fire, never in the machinery limiting how often.

Determinism and auditability. Cooldown durations are sampled from a per-participant seeded generator, so each participant's cue cadence is reproducible. Every session logs its allocation source (block-stratified table or parity fallback), random seed, and a per-policy validation row asserting that the run's parameters match the planned shared constants - making every session's configuration recoverable after the fact.

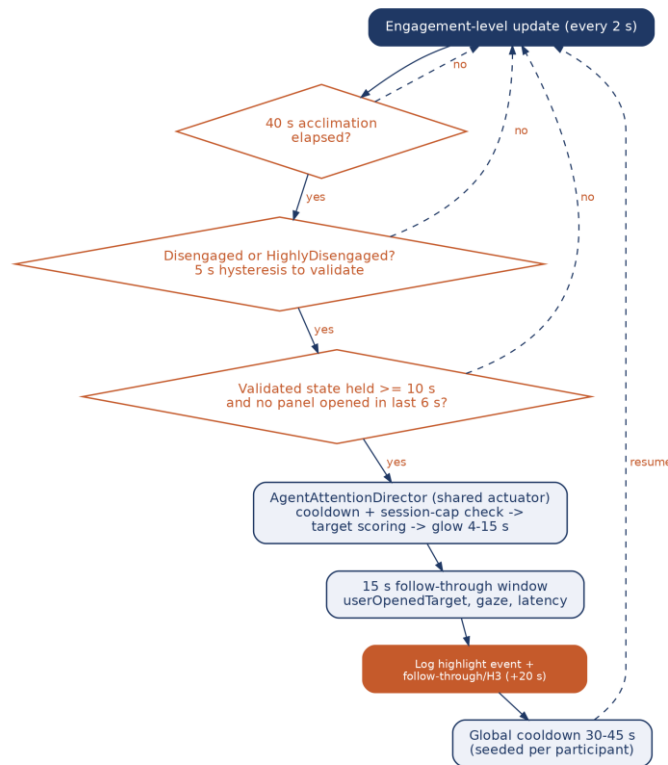


Figure 5. diagram of the Adaptive trigger policy.

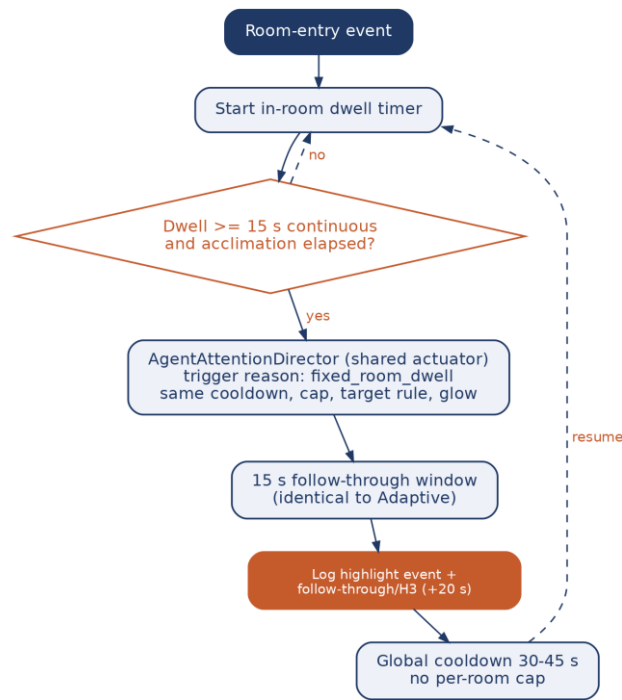
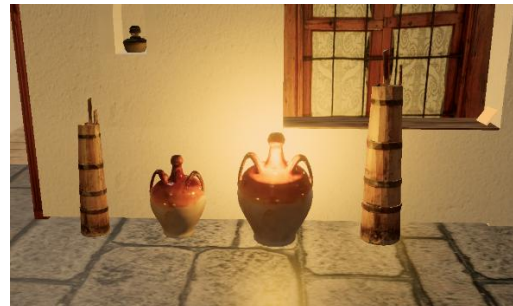


Figure 6. diagram of the Control (fixed-schedule) trigger policy.



A



B

Figure 7. The shared highlight cue: (A) an artefact in its normal state; (B) the same artefact during an active highlight - a pulsing glow with an optional spatialised chime. A full demonstration is available in the demo video in the project repository.

2.5 Telemetry and Data Persistence

All behavioural and event data flow through `FirestoreLogger`, which builds hierarchical `Firestore` paths of the form `users/{participantCode}/sessions/{sessionId}/{collection}/{docId}`, injects a server timestamp into every document, queues writes issued before initialisation, and blocks writes while tracking is invalid. More than twenty event and telemetry streams are populated per session. The analytically central records are `engagementStates` (periodic and transition-triggered classifier samples with all sensor scores), `agentInterventions`

(highlight-fire and highlightFollowThrough rows containing the target, response, latency, pre/post smoothed scores, and smoothedScoreDelta used for H3), objectViewingTime and readingTime, the raw sensor streams, contentAdaptation, sessionSummary, and studyValidation. User-level collections store demographics, condition allocation and provenance, and gamification progress. The schema was designed before recruitment so that every planned measure could be derived without post-hoc instrumentation.

2.6 Research Method

The final study is a between-subjects pilot with the trigger policy as the single independent variable, one session per participant. Everything else is held constant through shared constants: cue appearance, target-selection rule, global cooldown range, session cap, acclimation window, follow-through window, and post-cue measurement window. Participants are allocated by a block-stratified table (2:2 within each block of four) with a participant-code parity fallback, assigned automatically at session start and never disclosed. Three hypotheses were specified a priori: H1 - Adaptive participants spend more time in the museum (session duration); H2 - Adaptive participants accumulate more reading view time (panel views flagged wasLikelyRead); H3 - the participant-level mean of the per-highlight change in smoothed engagement between the pre-fire snapshot and the +20 s snapshot is greater under Adaptive. Because both conditions deliver highlights, H3 is the dose-controlled comparison that addresses RQ2. Secondary measures: highlight dose, follow-through, gaze follow-through, latency-to-interaction, distinct artefacts examined, engagement-level distribution, SUS usability, UES subjective engagement, and a pre/post cultural-knowledge test.

Each session followed the same protocol: informed consent and in-application demographics (age group, gender, nationality, computer skill, prior VR experience) plus baseline measures; headset fitting with IPD adjustment; on-screen navigation instructions during the 40-second acclimation; free exploration for up to about twenty minutes, ending at the exit portal; then the post-session questionnaire (SUS, UES, knowledge post-test) and a debrief. Participants were not told their condition, the hypotheses, or that artefacts would be highlighted, preserving natural behaviour. The study followed the institution's ethical guidelines and received approval before data collection. Analyses use Welch's t and Mann–Whitney U tests with Cohen's d, Hedges g, and 10,000-sample bootstrap 95% confidence intervals; given the pilot scale, interpretation emphasises effect size and direction over significance, and sessions enter each analysis according to the validity of the corresponding logs (staged inclusion, Section 5.1).

3. Development and Research Process

3.1 Client Collaboration and the Digital Twin

The project began as a collaboration with Sokol, a cultural-heritage partner from Tirana University, Albania, who acted as the client throughout. The first deliverable was a faithful digital twin. Artefact models came from three sources: photogrammetry, where photographs taken from several angles were processed by image-to-3D services into accurate but extremely heavy FBX meshes (often over a million polygons); online repositories, whose base models were adjusted to match specific items in the collection; and manual rebuilding in Blender - retopology and polygon reduction to clean low-poly meshes, followed by UV unwrapping, normal-map baking, and texture cleanup to recover the detail lost in reduction (Figure 8). Many artefacts needed several Blender sessions to reach an acceptable polygon count while remaining recognisable.

Client interaction was continuous rather than milestone-based. Recurring Zoom reviews with Sokol compared renders of each rebuilt artefact against reference photographs, verified display-case placement and room transitions against the real floor plan, reviewed the information-panel texts and their Albanian translations for cultural accuracy, and - when polygon and memory budgets forced the collection to be trimmed - decided which items were culturally indispensable. This supplied the local knowledge that turned a generic VR scene into a faithful reconstruction.

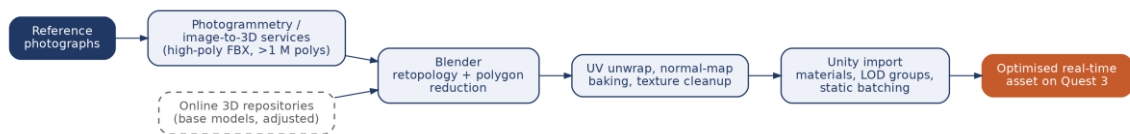


Figure 8. Asset pipeline from source photographs to optimised, real-time-ready models.

3.2 Platform Evolution and Tooling

The first Unity build targeted Android phones and dropped below ten frames per second in artefact-dense rooms. A profiling-driven optimisation pass (detailed in Section 4.1) restored a steady thirty frames per second - adequate for a phone prototype, but far below comfortable headset rates, which motivated the move to the Meta Quest 3 (Snapdragon XR2 Gen 2, Adreno 740, 2064×2208 per eye). The migration brought full 6DoF locomotion, spatial audio, hand tracking, and the XR Interaction Toolkit's ray-based interaction model, which in turn became the primary gaze channel of the engagement classifier.

Tool / technology	Role
Unity 6000.2.8f1 + URP	Engine; mobile-optimised rendering; Android/Quest build
XR Interaction Toolkit 3.x, Meta OpenXR, XR Hands	Locomotion, ray interactors, 6DoF tracking, hand-joint data
C# - Visual Studio 2022 / JetBrains Rider	Implementation and debugging with Unity integration
Blender 3.x/4.x + photogrammetry services	Asset pipeline: retopology, baking, texture work
Firebase Firestore + Anonymous Auth	Telemetry persistence; per-participant isolation
OpenAI GPT API	Exhibit-text generation (fixed settings in the final study)
Git / GitHub	Version control; branches for VR, Android, and WebGL builds

Tool / technology	Role
Meta Quest Developer Hub, Unity Profiler, Firebase Console	Sideloaded and device logs; frame analysis; live telemetry inspection
Zoom	Client reviews and remote collaboration

Table 3. Principal tools and technologies.

3.3 Research Iteration 1: Adaptive Content Pacing

The first research iteration implemented the engagement classifier together with an LLM feature that adapted exhibit-text length to the inferred engagement level: an engaged visitor received richer descriptions, a disengaged one shorter text, generated at runtime by GPT-3.5-turbo with hard caps by engagement level, a rolling best-of-window at request time, and a five-minute cache for latency and cost. A preliminary between-subjects study ($N = 16$; eight per group; Table 4) compared this adaptive pacing with a static baseline: the adaptive group showed substantially longer sessions, more reading time, and more content exposure, with large effect sizes (session duration $d = 1.38$; reading view time $d = 1.37$; words exposed $d = 1.36$), higher usability (SUS 84.7 vs 78.1), and a medium effect on perceived presence.

The work was accepted and presented at the SHAPEXR Workshop at ACM IUI 2026 in Limassol, Cyprus (July 2026) [1, 2].

Characteristic	Adaptive (n = 8)	Control (n = 8)
Prior VR experience	Yes: 5; No: 3	Yes: 4; No: 4
Controller gaming frequency	Frequently: 4; Occasionally: 2; Rarely: 2	Frequently: 1; Occasionally: 3; Rarely: 2; Never: 2
Museum interest, M (SD)	3.12 (1.25)	3.25 (1.49)
Visit likelihood, M (SD)	3.75 (1.04)	3.88 (1.36)

Table 4. Iteration 1 participant characteristics by condition ($N = 16$; aged 13-34; 15 men, 1 woman). Groups were balanced on museum interest ($U = 30.0$, $p = .872$) and visit likelihood ($U = 28.5$, $p = .743$).

After analysing the results, however, we concluded that text-length adaptation did not provide sufficiently meaningful insight into visitor engagement: it changed how much visitors read, but it confounded content with timing and said little about when an adaptive system should act at all. The behavioural gains could reflect longer texts simply taking longer to read. This critique motivated a redesign around a manipulation whose only degree of freedom is time.

3.4 Research Iteration 2: Engagement-Driven Highlighting

The second iteration pivoted from adapting what visitors read to guiding where they look (Figure 9). The redesign had four parts. First, confound removal: the avatar and speech-bubble surfaces were retired, and the LLM text settings were frozen to engagement-independent values identical in both conditions, leaving trigger timing as the single between-group difference. Second, a matched Control: a fixed-schedule policy was built over the same actuator, cooldown, and cap as the Adaptive policy - a control condition the first study lacked. Third, reproducibility: seeded cooldowns, automatic block-stratified allocation, and per-session parameter-validation rows. Fourth, verification: a functional test pass covering login and allocation, classifier behaviour (velocity gates, reading verification, tracking validation), both trigger policies (acclimation, cooldown, session cap), gamification, localisation, and Firebase data integrity across roughly forty scripted test cases. The pilot was then run with fourteen participants; because the manipulation changed fundamentally, the first iteration's data could not be reused.

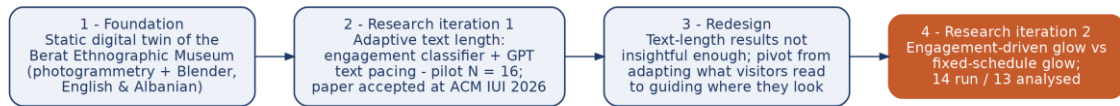


Figure 9. Evolution of the project from the foundation work through the two research iterations.

3.5 Verification and Testing

Verification combined scripted functional tests in the Unity editor with complete device-level runs on the Meta Quest 3. Approximately forty planned test cases covered login and condition allocation, the 40-second acclimation gate, classifier thresholds and velocity gates, reading verification and panel-occlusion correction, both trigger policies, the shared cooldown and session cap, target selection, gamification, localisation, participant-state isolation, and Firebase data integrity.

Each release candidate was smoke-tested from login through the exit portal, followed by inspection of the corresponding Firestore collections. After the head-tracker and reading-gate faults were discovered, regression testing added repeated back-to-back sessions without restarting the application, scene-transition checks, flat-signal detection, Adaptive-trigger verification under prolonged disengagement, and confirmation that the corrected build continued logging valid head and movement samples.

4. Challenges

4.1 Mobile VR Performance

Photogrammetry-quality assets and a standalone mobile GPU pull in opposite directions, and the first build made the tension concrete: under ten frames per second in rooms with several artefacts. Profiling attributed the cost to four causes, each fixed in turn. Draw-call overhead - every artefact initially carried its own material, generating hundreds of draw calls per frame; materials were merged and static batching enabled for immobile geometry. Texture memory - 4096×4096 photogrammetry textures were downscaled to 1024² or 512² according to artefact size and typical viewing distance, with mipmapping. Real-time lighting - the closed rooms were re-lit with pre-baked lightmaps, removing per-frame lighting cost entirely. Geometry - artefacts visible across rooms received level-of-detail groups switching to low-poly versions beyond two metres. Together with the Blender retopology work (Section 3.1), these brought the Android build to a steady 30 fps; the residual gap to comfortable headset rates was closed by moving to the Quest 3's substantially stronger XR2 Gen 2 platform.

4.2 Inferring Engagement Without Eye Tracking

The Quest 3 has no eye tracker, so engagement had to be estimated from behavioural proxies - the central analytical challenge of the platform and a known limitation of non-eye-tracked VR analytics [4]. Three design problems dominated. Weighting: the four channels differ in reliability and intent; controller-ray gaze is the only signal under deliberate user control, so it carries 40%, head stability and locomotion 25% each as passive corroboration, and hand proximity 10% since controllers make it partly redundant. Smoothing: a symmetric moving average, tested first, systematically rescued genuine disengagement - in internal traces, the vast majority of raw scores below 0.30 were smoothed back above the intervention threshold, so the classifier almost never saw sustained disengagement. The solution was asymmetric smoothing whose descent

coefficient is fifty percent faster than its ascent (0.525 vs 0.35): drops register promptly, upward flickers are dampened, and oscillation at level boundaries disappears. Motion states: a visitor striding between rooms scores 'stable head, moving legs' - which a naive fusion reads as moderately engaged; two velocity gates cap the level at Disengaged during fast locomotion without reading evidence, with a reading-grace suppression so that walking away from a just-read panel is not punished. All parameters were frozen before recruitment to keep the study honest.

4.3 Sensor Artefacts: the Panel-Occlusion Problem

Midway through development, engaged readers began registering as disengaged - the opposite of ground truth. The cause was geometric: opening a content panel places UI between the controller ray and the artefact's collider, so the raycast reports zero gaze precisely while the visitor is reading, the strongest engagement evidence the system has. Score-level patches (boosting the fused score during reading) would have double-counted the reading channel; instead, the fix was sensor-level: while a panel is open the gaze channel is floored at 0.85, and during sustained high-confidence reading at 0.70, restoring what an unobstructed sensor would have measured. Reading itself still enters the score only through its own 25% term. A complementary problem - the reading tracker producing false positives whenever the visitor lingered near text-bearing surfaces - was solved by content-anchored verification: reading evidence is zeroed unless a panel is actually open or was closed within the last fifteen seconds, and sub-two-second 'skipped opens' do not extend that window.

4.4 Isolating a Single Experimental Variable

The scientific value of the study rests on one claim: the only difference between groups is when highlights fire. Honouring that claim shaped the architecture. The actuator was extracted into a component shared by both policies, so target selection, glow, chime, dismissal, and measurement are byte-identical across conditions. Every cue parameter lives in shared constants consumed by both policies. Features that could differ between groups - the avatar guide, its speech bubble, adaptive text pacing - were disabled in both. Randomness was tamed: cooldown durations are drawn from a per-participant seeded generator, so cue cadence is reproducible; allocation is block-stratified with a deterministic parity fallback and is logged with its provenance; and each session writes a validation row asserting its parameters match the plan. The result is an auditable experiment in which any session can be reconstructed from its logs.

4.5 Mid-Study Implementation Faults

Two faults surfaced only under real participants. The head-tracker persistence bug: HeadTrackingAnalyzer survives scene loads as a persistent singleton, and after the login-to-museum transition it could retain a reference to the destroyed login-scene camera; head and movement telemetry then froze at constant values while everything else ran normally. It froze logging in three sessions before being diagnosed from the flat telemetry curves and fixed mid-study by re-acquiring the camera reference on scene load. The reading-protection gate: a guard designed to avoid interrupting readers was tuned so conservatively that one content-grazing participant - opening panels briefly but never reading - was disengaged for long stretches yet received no cues at all. Both faults raised the same methodological question: discard affected participants or salvage what is valid? We chose staged inclusion - classifying each session by the analyses for which its logs

remain valid and reporting the exact n per outcome - preserving statistical power where the data are sound while excluding only what is genuinely compromised. One session with wholly invalid logs was excluded outright; one participant who logged out mid-session was re-run under the same code, separated by session ID.

4.6 Shared Devices, State, and Localisation

Running many participants on one headset makes state isolation a correctness requirement: collectible-card and badge state is keyed per participant code in PlayerPrefs, all Firestore paths are rooted at the participant's user document, and the login flow detects a code change and resets local state, which the test plan verified with back-to-back sessions. Full bilingual support (English/Albanian) was maintained from the start through Unity string tables covering UI, room names, card titles, artefact panels, and badge descriptions - a continuous discipline, since every new feature added strings that needed translation and review by the client rather than a bolt-on at the end.

5. Results and Conclusions

5.1 Participants and Data Quality

Fourteen participants completed a study run; one Adaptive session was excluded because its logs were unusable, leaving thirteen analysed (seven Adaptive, six Control). In the analysed sample, ten participants (77%) were aged 25-34 and three (23%) were aged 18-24, with a balanced gender distribution across conditions (Table 5). Because validity depended on the required data stream, staged inclusion was applied: session-duration, reading, and content outcomes use $n = 7/6$ (Adaptive/Control); engagement-distribution uses $n = 6/4$; follow-through and H3 use $n = 5/6$; and gaze follow-through uses $n = 5/4$. This preserves valid observations while making the exact sample size for every outcome explicit.

Characteristic	Adaptive (n = 7)	Control (n = 6)
N (survey-valid)	7	6
Age group (years)	18–24: 1; 25–34: 6	18–24: 2; 25–34: 4
Gender	M: 3; F: 3; other: 1	M: 3; F: 3

Table 5. Participant characteristics by condition.

5.2 Outcomes

Primary outcomes. H1 and H2 were not supported: session duration (393 vs 400 s; $g = -0.05$; $U p = .943$) and reading/content-view time (104 vs 95 s; $g = 0.14$; $U p = .830$) showed no meaningful between-condition difference. H3 showed exploratory support: among sessions with at least one delivered highlight and valid pre/post fields ($n = 5/6$), mean per-session engagement recovery was positive under Adaptive and slightly negative under Control (+0.196 vs -0.038; Hedges $g = 1.79$; Welch $p = .026$; bootstrap 95% CI of the difference [0.09, 0.36]). A supplementary per-highlight-window analysis showed the same direction (+0.242 over 14 Adaptive windows vs -0.052 over 36 Control windows; Welch $p = .0001$), but the windows are nested within participants and are therefore reported only as exploratory support.

Metric	Adaptive M (SD)	Control M (SD)	p	Effect
Session duration (s)	393 (66)	400 (188)	$U = .943$	$g = -0.05$
Reading / content view time (s)	104 (44)	95 (66)	$U = .830$	$g = 0.14$
Content opens (count)	18.1 (7.9)	13.2 (8.9)	$U = .517$	$g = 0.55$
Highlight dose (count)	2.8 (1.3) ‡	6.0 (2.8)	$U = .064$	$g = -1.28$
Follow-through (acted on)	87% ‡	35%	Welch = .044	$g = 1.26$
Gaze follow-through	87%	55%	$U = .227$	$g = 0.75$
Latency to interaction (s)	5.6 (1.4)	2.8 (2.3)	$U = .233$	$g = 1.38$
H3 recovery / session (Δ)	0.196 (0.159) ‡	-0.038 (0.076)	Welch = .026	$g = 1.79$

Table 6. Behavioural outcomes by condition. M (SD); the p column identifies the reported test. Exact n (Adaptive/Control): duration, reading, content opens, and dose = 7/6; follow-through, latency, and H3 = 5/6; gaze = 5/4. Effect sizes are Hedges g . ‡ = bootstrap 95% CI of the difference excludes 0.

Secondary outcomes. Adaptive delivered fewer highlights (2.8 vs 6.0 per session; $n = 7/6$; $g = -1.28$; $U p = .064$; bootstrap CI excluding zero), yet participants acted on a much larger proportion of its cues (87% vs 35%; $n = 5/6$; $g = 1.26$; Welch $p = .044$; CI excluding zero). The cued object was opened after 87% of Adaptive highlights versus 26% of Control highlights. Gaze follow-through, restricted to sessions with live tracking, was directionally higher under Adaptive (87% vs 55%; $n = 5/4$) but not statistically reliable. Interaction

latency was longer under Adaptive (5.6 vs 2.8 s), an important counter-finding consistent with cues reaching visitors while they were disengaged and slower to re-orient. Control was rated higher on SUS (92 vs 81; Welch $p = .086$), although both means exceeded the conventional acceptable-usability benchmark; UES engagement, knowledge gain, and baseline measures were comparable.

Qualitative feedback matched the behavioural pattern. Control participants disliked the fixed cues: one called the experience “chilling,” another explained, “*Sometimes it bothered me rather than helping, because I was already engaged and interacting with other objects and it kept drawing my attention away.*” An Adaptive participant instead said, “*I wanted more highlights, because sometimes I was just wandering.*”

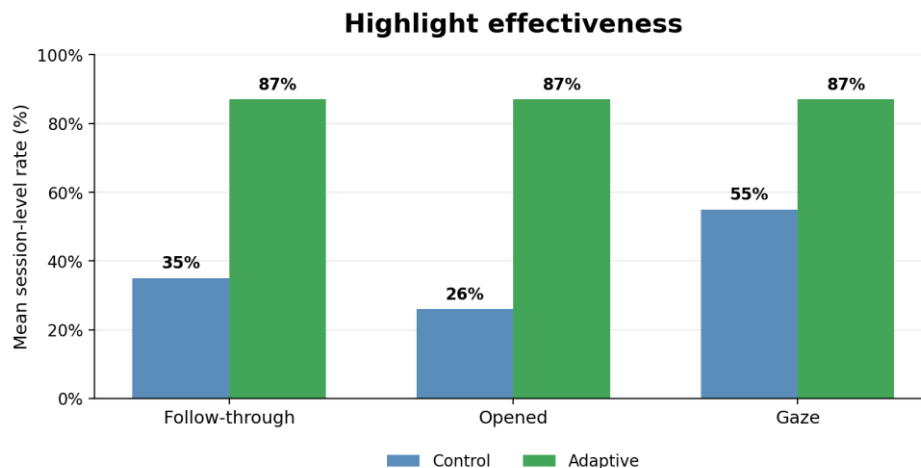


Figure 10. Highlight effectiveness by condition: follow-through, opened, and gaze (live-tracker sessions). Exploratory.

5.3 Conclusions

Regarding RQ1, the policies did not differ meaningfully in total session duration or reading time; the difference appeared around the moments in which cues were delivered. Adaptive produced fewer than half as many highlights, yet participants acted on 87% of them compared with 35% under Control. Regarding RQ2, this pattern is directionally more consistent with timing than with dose: engagement increased after Adaptive cues and remained flat or decreased after fixed-dwell cues, despite the lower Adaptive cue count. Within this exploratory pilot, engagement-driven timing therefore showed a more efficient follow-through and recovery pattern than fixed scheduling, rather than a general increase in time spent in the museum.

Two results temper the picture. Control's higher SUS and Adaptive's longer interaction latency may reflect the fact that Adaptive cues deliberately arrive during disengagement, when visitors are slower to respond and may be less receptive to interruption. The principal validity threats are the small sample, outcome-specific missing data, and reliance on the same heuristic engagement signal both to trigger Adaptive cues and to calculate H3. An exploratory no-cue comparison suggested that cued recovery exceeded the clean Control-only natural-rebound baseline but not the more inclusive baseline; H3 should therefore be interpreted as recovery associated with the policy, not as a causal estimate of cue-driven recovery. A larger confirmatory study should use a within-subject design, independently validated engagement measures, and time-locked no-intervention disengagement epochs.

Criterion	Target	Outcome	Notes
Minimum sample size	$N \geq 20$ (10 per condition)	Not met	14 run, 13 analysed; results treated as exploratory
SUS score (both conditions)	≥ 70 (acceptable usability)	Met	Adaptive 81, Control 92
Follow-through (Adaptive)	$\geq 50\%$ of highlights opened	Met	87%
Firebase data completeness	$\geq 95\%$ complete summaries	Partially met	13 of 14 sessions (93%) analysable
Tracking-validity rate	$\geq 90\%$ of each session valid	Partially met	Tracker bug froze logging in 3 sessions; fixed mid-study; staged inclusion applied
H3 (engagement recovery)	Adaptive > Control	Direction met (exploratory)	+0.196 vs -0.038; $g = 1.79$; CI excludes zero; not a causal estimate

Table 7. A-priori success criteria and observed outcomes.

6. Lessons Learned

What worked well. Freezing every experimental constant before recruitment, seeding the cooldowns, and logging a per-session validation row proved their worth the moment mid-study faults appeared: every session's exact configuration was recoverable, and staged inclusion could be applied honestly instead of discarding participants wholesale. Building both policies over one shared actuator is what makes the isolation of trigger timing credible, and the decision to design the full Firestore schema before the study meant every planned measure could be computed from the logs with no post-hoc instrumentation. On process, the two-iteration structure - run a study, analyse honestly, redesign - turned a modest first result into a sharper second question, and sustained client reviews kept the reconstruction culturally faithful rather than generic.

What we would change. The head-tracker persistence bug cost three sessions of head and movement data; a full end-to-end telemetry rehearsal on the device - several complete sessions across real scene transitions, verifying every Firestore collection afterwards - would have caught it before recruitment and is the single clearest process improvement. The reading-protection gate taught a related lesson: guards that interact with the experimental manipulation must be stress-tested against edge-case behaviour profiles (here, a visitor who grazes content without reading). We would also recruit to the planned $N = 20$ before analysing, budget explicit time between iterations for instrument validation rather than moving straight to the next study, and capture UI screenshots systematically during development for documentation.

Objectives and success criteria. Measured against the Phase A plan, the engineering objectives were achieved: the museum was reconstructed and localised; the closed-loop classifier runs on-device with sub-millisecond inference; both trigger policies use one shared, auditable cue actuator; and the telemetry pipeline supported the planned analyses. The research objectives were achieved in part (Table 7): recruitment stopped below the target sample and two data-quality criteria were only partially met, while the H3 direction and follow-through target showed exploratory support and both conditions exceeded the usability target. The project nevertheless converted an open interaction-design question - when should a VR museum intervene? - into a controlled and reproducible pilot, produced two workshop publications, and delivered the infrastructure required for a stronger confirmatory study.

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Appendix A - User Guide

This guide provides operating instructions for the Adaptive VR Museum, organised by use-case scenario. It covers the normal successful flow for the system's two user roles: the visitor exploring the museum and the experimenter conducting a study session. The introductory controls panel is the application's built-in User Help mechanism and is displayed at the start of every visit. All screens and content are available in English and Albanian (Shqip); the selected language applies to panels, room names, cards, badges, and the achievements screen.

A.1 Controls at a Glance

Action	Control
Move	Left thumbstick (continuous locomotion)
Turn	Right thumbstick (snap turn)
Point at an object or button	Aim the controller - a visible ray extends from it
Select / open / press	Point the ray and pull the trigger
Type	Point at a text field, pull the trigger, then use the on-screen keyboard
Close a panel	Point at its close button and pull the trigger, or walk away

Table A.1. Controller mapping.

A.2 Scenario 1 - Starting a Visit

1. Put on the headset and adjust the strap until the text on screen is sharp; take one controller in each hand.
2. On the login screen, select your language (English / Shqip) with the toggle in the corner.
3. Point the ray at the participant-code field, pull the trigger, and type the code you were given; press Submit.
4. First visit only: a demographics panel appears with five questions - age range, gender, nationality, computer skill, and prior VR experience - answered by pointing and selecting. The Continue button activates once all five are answered.
5. The museum scene loads and the built-in User Help panel explains movement, turning, pointing, selection, and panel interaction. Read it at your own pace, then press Start. Returning participant codes skip the demographics form and arrive directly at this panel.

A.3 Scenario 2 - Exploring Rooms and Artefacts

6. Walk through the six rooms in any order and at your own pace; there is no required route. The experimenter will tell you when the planned study period has ended if you have not already used the exit portal.
7. On your first entry to each room, a popup shows the room's name and a short cultural description, then closes by itself.
8. To examine an artefact, point the ray at it and pull the trigger: an information panel opens beside it with the artefact's name and description. Read for as long as you like.
9. Close the panel with its close button, or simply walk on - each examined artefact adds ten points to your score.



Figure A.1. In-headset view of the kitchen room; the glowing vessel on the right is an active highlight cue.

A.4 Scenario 3 - Responding to a Highlight

10. From time to time, a nearby artefact you have not yet examined may begin to glow softly, sometimes with a brief chime (Figure A.1).
11. The glow is an invitation, not an instruction: walk over and open the artefact's panel if you are curious, or ignore it and continue - it fades by itself after a few seconds.
12. Looking directly at the glowing artefact for a moment also dismisses the glow.

A.5 Scenario 4 - Collecting Cards and Badges

13. Eighteen cards are hidden across the museum, three per room. Walking close to one opens a discovery panel with an illustration and the story behind it; press the confirmation button to collect it (one hundred points each).
14. Badges unlock automatically as you progress - at one, three, six, nine, twelve, and eighteen cards, for collecting all three cards in a room, and a master badge for completing everything. Each unlock announces itself with a slide-in notification; collecting everything also earns a one-thousand-point bonus.

A.6 Scenario 5 - Ending the Visit

15. When you are ready to finish, walk through the exit portal.
16. The achievements screen appears automatically: your badges, collected cards, time spent in each room, and total score, in your chosen language. Take your time reviewing it - this ends the visit.

A.7 Experimenter Workflow

Preparing a session. Charge the headset fully (connect to power between back-to-back sessions); confirm Wi-Fi connectivity, since telemetry requires it; launch the application once and check it reaches the login screen; take the next participant code from the allocation table; and confirm in the Firebase Console that the code has no stale documents. Clear a guardian boundary of at least 2 m × 2 m.

Running a session. Seat or stand the participant inside the boundary, assist with fitting, and adjust the inter-pupillary distance until the login text is sharp. Have them log in with their code and complete the demographics - the condition is assigned automatically and is never shown. From then on, monitor passively: do not point out rooms or artefacts, and

if asked for help, only restate the navigation controls. The 40-second acclimation runs by itself.

Closing a session. The session ends when the participant exits through the portal (or after about twenty minutes). Let them view the achievements screen, help remove the headset, and administer the post-session questionnaire (SUS, UES, knowledge post-test). Debrief them about the study and the two conditions, record the participant code and any observations, and verify in the Firebase Console that the session-summary document is present and complete before the next participant.

Appendix B - Maintenance Guide

This guide supports future maintenance, extension, and continued use of the project. It describes the operating environment, repository layout, installation and deployment of the software developed in this project, the configuration surface, the data model, and common extension points. The appendix is self-contained for routine maintenance tasks.

B.1 System Overview for Maintainers

All logic runs on-device in a single Unity application; there is no project-owned server. Three scenes build in order - LoginScene → FullMuseum → AchievementsScene. At launch, FirebaseManager signs in anonymously and initialises Firestore; ExperimentConditionManager derives the condition from the participant code and enables exactly one trigger policy; SessionPhaseManager starts the session clock and enforces the acclimation gate. The four sensor singletons feed EngagementClassifier, which emits an engagement level every two seconds; the active policy decides when to call the shared AgentAttentionDirector, which selects a target and drives GuidedHighlight; and every subsystem writes through FirebaseLogger. The package diagram in Figure 2 (Section 2.2) shows these relationships.

B.2 Operating Environment

Component	Requirement
Engine	Unity 6000.2.8f1 with Universal Render Pipeline (URP)
XR packages	XR Interaction Toolkit 3.x; Meta OpenXR; XR Hands; Unity Localization; Addressables
Target device	Meta Quest 3 - Android, ARM64, IL2CPP scripting backend; Developer Mode enabled
Backend services	Firebase project with Firestore and Anonymous Authentication enabled
LLM service	OpenAI API account and key (exhibit-text generation)
Developer workstation	Windows with Unity Hub, Meta Quest Developer Hub (or adb), Git; Visual Studio 2022 or Rider
Network	Wi-Fi for the headset - required at runtime for Firestore and the OpenAI API

Table B.1. Operating environment and infrastructure.

B.3 Repository Layout

Location	Contents
Assets/Scripts	Sensor and classifier components (EngagementClassifier, ControllerGazeTracker, HeadTrackingAnalyzer, HandProximityTracker, ReadingBehaviorTracker), SessionPhaseManager, ExperimentConditionManager, StudyConstants, and LLMAdaptiveContentManager

Location	Contents
Assets/Scripts/Agent	Intervention system: MuseumAgentController, NonAdaptiveHighlightPolicy, AgentAttentionDirector, GuidedHighlight, AgentFirebaseLogger
Assets/Scripts/Firebase	FirebaseLogger, FirebaseManager (auth, demographics, write API)
Assets/Scripts/Gamification	HiddenCard, BadgeManager, ScoreManager, AchievementsManager, notification UI
Assets/Scenes	LoginScene, FullMuseum (main scene; serialized study parameters), AchievementsScene
Assets/Locales	English and Albanian string tables (Unity Localization)
Assets/StreamingAssets	openai_config.json (API key - never committed)
Assets/3Dobjects, Geometries, Textures, Materials	Museum geometry and optimised artefact assets

Table B.2. Repository layout (paths under the Unity project root).

B.4 Installation and First Build

1. Clone the repository and open the project folder with Unity Hub; install Unity 6000.2.8f1 when prompted (the exact version matters - URP and XR package versions are pinned to it). Let the package import finish completely on first open.
2. Firebase: create or select a Firebase project with Firestore and Anonymous Authentication enabled, register the Unity Android application ID, download google-services.json, and place the file unchanged in the Unity project Assets folder. FirebaseManager performs anonymous sign-in at launch; no source-code change is required.
3. OpenAI: create Assets/StreamingAssets/openai_config.json locally with the content {"apiKey": "sk-..."} . This file is required by the Quest build for live exhibit-text generation and must never be committed. Rotate the key immediately if it is exposed.
4. Switch the build target to Android; verify Texture Compression ASTC, scripting backend IL2CPP, target architecture ARM64, and the OpenXR plug-in with the Meta Quest feature set enabled.
5. Confirm the scene list order in Build Settings: LoginScene, FullMuseum, AchievementsScene - the flow depends on this order.
6. Build the APK. Enable Developer Mode on the headset, connect it by USB, approve the connection, and install with Meta Quest Developer Hub (Install APK) or 'adb install -r <apk>'.
7. Smoke test: launch on the headset, log in with a test code, confirm the demographics panel appears, enter the museum, open one artefact panel, and verify in the Firebase Console that documents appear under users/<code>/sessions/... within a minute.

B.5 Configuration Reference

Experiment behaviour is controlled by a combination of constants in StudyConstants.cs and inspector-exposed fields serialized in the FullMuseum scene. Table B.3 lists the principal parameters and their effective final-build values. Before any new study, verify the values against the exact build and inspect the per-session studyValidation row, which records the runtime configuration used for that session.

Parameter	Script · field	Meaning
Classifier tick	EngagementClassifier.updateInterval (2 s)	Inference cadence

Parameter	Script · field	Meaning
Physical/reading split	physicalSignalWeight / readingSignalWeight (0.75/0.25)	Base fusion split
Sensor weights	gazeWeight / headStabilityWeight / movementWeight / handProximityWeight (0.40/0.25/0.25/0.10)	Physical sub-score
Level thresholds	highlyEngaged/engaged/neutral/disengaged Threshold (0.75/0.55/0.35/0.20)	Score → level mapping
Smoothing	levelTransitionSpeed (0.35; ×1.5 descent)	Asymmetric EMA
Velocity gates	slowMovementThreshold / veryFastThreshold (0.8/1.5 m/s)	Cap level while moving
Acclimation window	SessionPhaseManager (40 s)	No cues before this
Adaptive trigger	MuseumAgentController - stateChangeHysteresis; disengaged/highlyDisengaged EscalationTime; minSecondsSinceLastContentOpen	Hysteresis, hold times, content-recency guard
Control trigger	NonAdaptiveHighlightPolicy.fixedRoomDwellBeforeHighlight (15 s)	Fixed-schedule dwell
Cue budget	AgentAttentionDirector - globalHighlightCooldownMin/Max; maxHighlightsPerSession (15)	Shared cooldown range and session cap
Targeting	min/maxTargetDistance (0.6–6.0 m); notYetInteractedBonus (+100); roomMatchBonus (+50); distancePenalty (–5/m)	Candidate range and scoring
Glow lifecycle	highlightMin/MaxDuration; targetLookDotThreshold (0.85); gazeToDismissSeconds (1.5 s)	Cue duration and gaze dismissal
Measurement windows	followThroughWindow (15 s); intervention-window delay (20 s)	Follow-through and H3 snapshot
Study mode	ExperimentConditionManager.experimentHighlightOnlyMode	Master switch disabling avatar / speech bubble / adaptive text

Table B.3. Principal configuration parameters (serialized in the FullMuseum scene).

B.6 Firestore Data Model

Two path patterns are used: per-session data at `users/{participantCode}/sessions/{sessionId}/{collection}/{docId}`, and per-user data at `users/{participantCode}/{collection}/{docId}`. The participant code is the user document identifier, Anonymous Authentication supplies the Firebase token, and a server timestamp is added to every document. The authoritative H3 fields are stored in `agentInterventions` follow-through records, including the pre/post smoothed scores and `smoothedScoreDelta`.

Collection (session-level)	Written by · key fields
<code>engagementStates</code>	EngagementClassifier · level, smoothed/raw score, four sensor scores, readingScore, velocity, trackingValid, logType (periodic transition)
<code>agentInterventions</code>	AgentFirebaseLogger · highlight-fire and highlightFollowThrough records: triggerReason, target, condition, userOpenedTarget, userGazedAtTarget, latencyToInteraction, pre/postSmoothedScore, smoothedScoreDelta - authoritative basis for H3
<code>contentAdaptation</code>	LLMAdaptiveContentManager · model, prompt, generated response, success flag, and response time; settings held engagement-independent in the final study
<code>objectViewingTime</code> / <code>readingTime</code>	InteractiveObject / ReadingBehaviorTracker · object id, viewDuration, wasLikelyRead, wordCount, confidence
<code>controllerGaze</code> / <code>headTracking</code> / <code>movement</code> / <code>handTracking</code>	Sensor trackers · raw per-sensor telemetry streams
<code>sessionPhase</code> / <code>sessionSummary</code> / <code>studyValidation</code>	SessionPhaseManager / EndOfSessionSummary / validation logger · phase transitions, duration, engagement distribution, score, condition, allocation source, random seed, and effective study constants

Table B.4. Session-level collections. User-level collections hold `experimentInfo` (condition, allocation source, seed), demographics, and progress (score, cards, badges, room stats).

B.7 Extension Points

Adding a trigger policy. Implement a `MonoBehaviour` that decides when to fire and delegates to `AgentAttentionDirector.TryHighlight()`; never draw the glow directly, so target selection, cooldowns, and measurement stay identical across policies. Register the policy with `ExperimentConditionManager` so allocation can enable it, and give it a distinct `triggerReason` string for the logs.

Adding an artefact or card. Duplicate an existing interactive-artefact prefab (collider + `InteractiveObject` + `GuidedHighlight` child) or `HiddenCard` trigger zone, place it in `FullMuseum`, and add its name and description keys to both string tables. Badge milestones and the three-cards-per-room logic pick up new cards automatically only if room counts stay consistent - adjust `BadgeManager` thresholds otherwise.

Adding a language. Add a locale in the Unity Localization settings, provide translations for every string-table key (UI, rooms, artefacts, cards, badges), and it appears in the login-screen selector without code changes.

B.8 Troubleshooting

Issue	Likely cause	Resolution
Login screen does not respond	Focus not on the text field	Point the controller ray at the field and pull the trigger
Firebase error on launch	No Wi-Fi, or wrong google-services.json	Check headset Wi-Fi and the Firebase configuration; relaunch
Exhibit text shows the fallback message	Missing or invalid OpenAI key	Verify <code>Assets/StreamingAssets/openai_config.json</code> , confirm that the key is valid, and inspect the device log for the key-resolution or API-response error
Head/movement telemetry frozen at constant values	Camera reference lost across a scene transition	Fixed in the current build; if it recurs, verify <code>HeadTrackingAnalyzer</code> re-acquires the camera on scene load
Highlight glow never appears	Policy disabled, acclimation still active, or prefab reference missing	Confirm the condition flags, wait out the 40 s acclimation, check the highlight prefab and device logs
Very low frame rate	Thermal throttling from a prior session	Let the headset cool between sessions
Session summary missing	Session not ended through the portal	Check for partial documents in the Console; mark the session for exclusion
Wrong condition assigned	Participant code mistyped	Codes map deterministically (block table / parity); verify the code and re-run under a fresh code

Table B.5. Common issues and resolutions.